

Automated Video to Spoken Summary Pipeline

Concise multi-tool workflow for production-ready summaries

End-to-end flow: MoviePy, Whisper, chunking, OpenAI summarization, gTTS synthesis



GAGAN



Prerequisites for Video Summarization Implementation

Libraries, models, and environment checklist



1 Python and environment

Python 3.8 to 3.11; use virtualenv or conda; update pip



2 Core libraries

moviepy, openai, gtts, torch, whisper (OpenAI Whisper or openai/whisper repo)



3 API keys and credentials

OpenAI API key for summarization; store via env vars or secrets manager



4 Hardware and performance

CPU-only workable; CUDA NVIDIA GPU with 8GB+ VRAM recommended



5 Audio toolchain and codecs

ffmpeg required by moviepy; ensure mp3, wav, aac codec support



Work Flow of the Model Pipeline

Procedure, code outline, and practical tips for speech-ready audio

Uploading Video and Extract audio

Upload Video to Convert Video to Audio Using Moviepy Model



Splitting into Chunks for Summarization

Breaking the Large Texts into smaller Chunks.



Conversion of Summary to Audio

Conversion of the Summary to Audio using Gtts Model



Extract Texts

Convert Audio to texts using Whisper Model



Summarization Using Open Ai-API Key

Generating a Meaningful Summary using Open AI by passing Chunks

Chunking Strategies for Long Transcripts: Approaches and Recommendations

Practical methods to preserve context and enable accurate summarization



1 Chunking methods



- ♦ Sentence-boundary: break at sentence ends, 20 to 50 token overlap
- ♦ Semantic segmentation: use embeddings to detect topic shifts
- ♦ Chapter or speaker detection: timestamps, silence, diarization

2 Practical recommendation



- ♦ Recommended chunk size: about 600 to 1200 tokens
- ♦ Overlap: 10 to 20 percent for OpenAI summarization
- ♦ Before/after example: 30-minute blob to six 5-minute coherent chunks
- ♦ Preserve context: keep sentence boundaries and small overlap

Step 4: Chunk Summaries with OpenAI API

Prompts, parameters, and merging strategy for deterministic summaries



API call pattern: send chunk text with system and user prompt plus metadata



Prompt template: summarize into 5 concise bullets, each 12 to 18 words



Context across chunks: summaries then synthesize globally



Merging strategy: concatenate summaries, dedupe, then output



Convert Summaries to Speech with gTTS

Procedure, voice options, file formats, and delivery tradeoffs



1

Prepare
Summary

2

Calling the gTTS
Model Library

3

Export Summary
Texts for
Conversion

4

Audio Saved

5

Export Audio